

requirement of accuracy in the whole seismic frequency zone, which leads to obvious numerical dispersion in wavefield propagation. To enhance the numerical solution accuracy, some scholars have proposed many ways to optimize the coefficients of the SFD operators, such as time-space domain dispersion-relation-based method (Liu and Sen 2011), simulated annealing algorithm (Zhang and Yao 2013), least-squares method (Yang et al., 2014a; Liu 2014) and so on. We (Yang et al., 2015) also proposed optimal implicit staggered-grid finite-difference schemes by using the sampling approximation. But, these optimization methods are only based on the standard staggered-grid format, and improve the numerical accuracy for the SSFD method. To decrease the numerical dispersion error in the RSFD method, it is extremely necessary to develop the optimization methods for the rotated staggered-grid format. We (Yang et al., 2014b) reported on the initial optimized formulas based on least-squares for rotated staggered-grid, but it could not consider the near zero-wavenumber constraint condition during the optimization process. In this paper, we explore the sampling approximation method and the least-squares method separately, and further develop them into the optimal FD schemes on rotated staggered-grid for elastic wave modeling in TTI media.

The structure of this paper is as follows: First, we propose the combination of the RSFD and the global optimization algorithms to derive the spatial difference coefficients, which are based on the sampling approximation (SA) method and the least-squares (LS) method, respectively. Then, the numerical accuracy of these RSFD schemes is analyzed according to the dispersion relation. We also use the optimal RSFD coefficients to solve the elastic wave equations for TTI media, and perform numerical modeling to examine the effectiveness of the new methods. Meanwhile, the modeling accuracy of the SA-based RSFD scheme and the LS-based RSFD scheme is compared with that of the TE-based RSFD scheme. Finally, we draw conclusions.

2. Theory of solving elastic wave equations with rotated staggered-grid finite-difference method

The elastic wave equations in 2D TTI media are (Winterstein 1990)

$$\frac{\partial v_x}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xx}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{xz}}{\partial z}, \quad (1a)$$

$$\frac{\partial v_y}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xy}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{yz}}{\partial z}, \quad (1b)$$

$$\frac{\partial v_z}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xz}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{zz}}{\partial z}, \quad (1c)$$

Besides, the transformation matrix R is expressed as (Liu et al., 1998)

$$R = \begin{bmatrix} \cos^2 \varphi & \sin^2 \varphi \cos^2 \theta & \sin^2 \varphi \sin^2 \theta & -\sin^2 \varphi \sin 2\theta & \sin 2\varphi \sin \theta & -\sin 2\varphi \cos \theta \\ \sin^2 \varphi & \cos^2 \varphi \cos^2 \theta & \cos^2 \varphi \sin^2 \theta & -\cos^2 \varphi \sin 2\theta & -\sin 2\varphi \sin \theta & \sin 2\varphi \cos \theta \\ 0 & \sin^2 \theta & \cos^2 \theta & \sin 2\theta & 0 & 0 \\ 0 & \frac{1}{2} \cos \varphi \sin 2\theta & -\frac{1}{2} \cos \varphi \sin 2\theta & \cos \varphi \cos 2\theta & \sin \varphi \cos \theta & \sin \varphi \sin \theta \\ 0 & -\frac{1}{2} \sin \varphi \sin 2\theta & \frac{1}{2} \sin \varphi \sin 2\theta & -\sin \varphi \cos 2\theta & \cos \varphi \cos \theta & \cos \varphi \sin \theta \\ \frac{1}{2} \sin 2\varphi & -\frac{1}{2} \sin 2\varphi \cos^2 \theta & -\frac{1}{2} \sin 2\varphi \sin^2 \theta & \frac{1}{2} \sin 2\varphi \sin 2\theta & -\cos 2\varphi \sin \theta & \cos 2\varphi \cos \theta \end{bmatrix}, \quad (4)$$

where φ and θ represent the azimuth angel and the tilt angel, respectively.

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial t} = & d_{11} \frac{\partial v_x}{\partial x} + d_{13} \frac{\partial v_z}{\partial z} + d_{14} \frac{\partial v_y}{\partial z} + d_{15} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \\ & + d_{16} \frac{\partial v_y}{\partial x}, \end{aligned} \quad (1d)$$

$$\begin{aligned} \frac{\partial \sigma_{zz}}{\partial t} = & d_{13} \frac{\partial v_x}{\partial x} + d_{33} \frac{\partial v_z}{\partial z} + d_{34} \frac{\partial v_y}{\partial z} + d_{35} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \\ & + d_{36} \frac{\partial v_y}{\partial x}, \end{aligned} \quad (1e)$$

$$\begin{aligned} \frac{\partial \sigma_{yz}}{\partial t} = & d_{14} \frac{\partial v_x}{\partial x} + d_{34} \frac{\partial v_z}{\partial z} + d_{44} \frac{\partial v_y}{\partial z} + d_{45} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \\ & + d_{46} \frac{\partial v_y}{\partial x}, \end{aligned} \quad (1f)$$

$$\begin{aligned} \frac{\partial \sigma_{xz}}{\partial t} = & d_{15} \frac{\partial v_x}{\partial x} + d_{35} \frac{\partial v_z}{\partial z} + d_{45} \frac{\partial v_y}{\partial z} + d_{55} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \\ & + d_{56} \frac{\partial v_y}{\partial x}, \end{aligned} \quad (1g)$$

$$\begin{aligned} \frac{\partial \sigma_{xy}}{\partial t} = & d_{16} \frac{\partial v_x}{\partial x} + d_{36} \frac{\partial v_z}{\partial z} + d_{46} \frac{\partial v_y}{\partial z} + d_{56} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) \\ & + d_{66} \frac{\partial v_y}{\partial x}, \end{aligned} \quad (1h)$$

where t is the time, x and z are the space coordinates, ρ is the density, (v_x, v_y, v_z) is the velocity vector, $(\sigma_{xx}, \sigma_{zz})$ is the normal stress vector, $(\sigma_{xy}, \sigma_{xz}, \sigma_{yz})$ is the shear stress vector, and $D = [d_{ij}]_{6 \times 6}$ represents the elastic parameter matrix under the observation coordinate system.

D can be computed by another elastic parameter matrix $C = [c_{ij}]_{6 \times 6}$ which is under the natural coordinate system and the transformation matrix R (Winterstein 1990)

$$D = RCR^T. \quad (2)$$

The form of the elastic parameter matrix $C = [c_{ij}]_{6 \times 6}$ is

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{bmatrix}, \quad (3)$$

with $c_{66} = \frac{1}{2}(c_{11} - c_{12})$.

Yang, Yan & Liu (2015) *J. Appl. Geophys.* 122:40–52 — Eq 3 (p. 41)

The elastic parameter matrix $C = [c_{ij}]_{6 \times 6}$ in the natural (VTI-symmetric) coordinate system:

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{bmatrix}, \quad (3)$$

with the VTI constraint (paper p. 41):

$$c_{66} = \frac{1}{2}(c_{11} - c_{12}).$$

Five independent VTI parameters: $c_{11}, c_{12}, c_{13}, c_{33}, c_{44}$ (with c_{66} derived). This is the natural-axis Voigt stiffness BEFORE Bond-rotation to the observation coordinate system. The observation-frame stiffness D is $D = RCR^T$ (Eq 2; R defined in Eq 4).